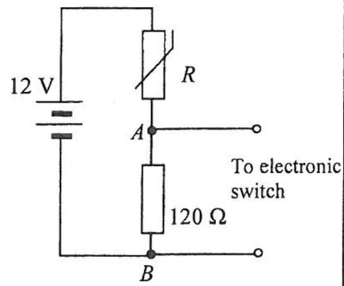
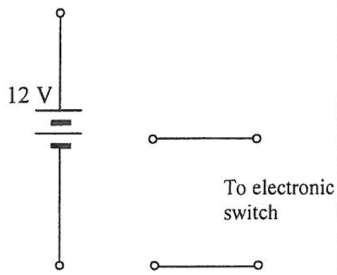
(c) (i) The potential difference V_{AB} is used to drive an electronic switch connected across AB to turn on a fan if temperature rises above a certain value such that V_{AB} is 6.0 V or above. Using the information provided in the graph, find the minimum temperature needed to keep the fan on. Show your working. (2 marks)



(ii) Without using additional components, complete the new circuit diagram below to illustrate how the circuit can be modified to turn on a heating device when temperature falls below a certain value. Explain the action of the circuit. No calculation is required. (3 marks)



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11. Figure 11.1 shows two identical small metal spheres X and Y suspended by insulating threads of the same length. Each sphere has a mass of 1.0×10^{-5} kg and each carries a positive charge of 3.1 nC (1 nC = 10^{-9} C). The separation d of the spheres is 10 cm. The size of the spheres is negligible compared with their separation, therefore they can be treated as point charges.

Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}\ \mathrm{N\ m^2\ C^{-2}}$. ($g = 9.81\ \mathrm{m\ s^{-2}}$)

Figure 11.1

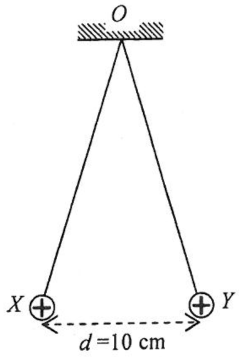

Diagram NOT drawn to scale

(a) Find the angle between the threads. (3 marks)

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(i) State which reading(s), V_m, I_m or both, do(es) **NOT** give the *true voltage* across the resistor and/or the *true current* passing through the resistor. Hence write down an equation relating R_A, R_m and R.
(2 marks)

(ii) Find the percentage error associated with R_m when measuring the resistance of this resistor.
Given: $R_V = 10\ \text{k}\Omega$, $R_A = 1\ \Omega$ and $R = 10\ \Omega$.
(2 marks)

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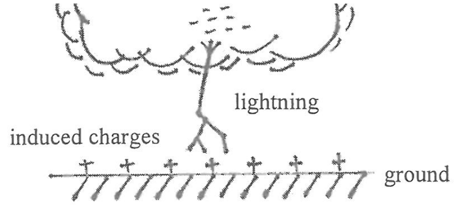

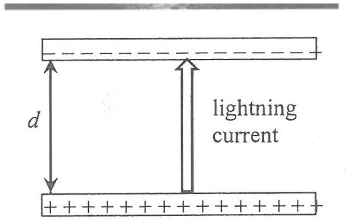

Lightning occurs when charges accumulate in the clouds to such an extent that the electric field in the atmosphere is strong enough to cause the air to lose its insulating properties. The threshold electric field for 'breakdown' to occur is about 3×10^5 V m^{-1} above which electrons or ions in the atmosphere can pass through the air between clouds and the ground or between clouds and clouds. The peak current of a typical lightning bolt can reach about 30000 A. How the charges are separated and accumulated in the clouds is not fully understood yet. In most cases, negative charges are at the base of the cloud and positive charges are induced on the ground.

(a) (i) What is the meaning of 'breakdown' in the passage ? (1 mark)

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*(ii) The thundercloud's base and the ground can be modeled as two parallel plates with opposite charges. If the negative charges distributed at the cloud's base are about $d = 2$ km from the ground, find the potential difference between the cloud and the ground when the electric field in the atmosphere just reaches the threshold of 'breakdown'. (2 marks)

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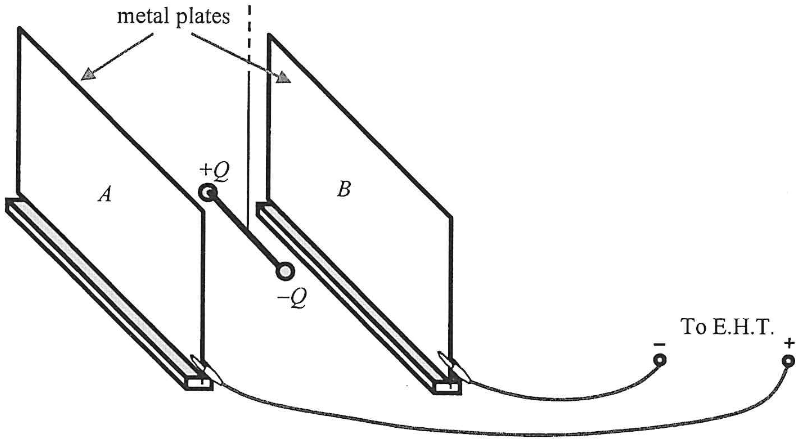
(c) Calculate the power delivered by the battery and show that the equivalent resistance of the circuit is slightly less than 1 Ω in this setting. (4 marks)

(d) Based on your answer in (c), explain whether a fuse rating of 15 A is suitable for this circuit or not. (2 marks)

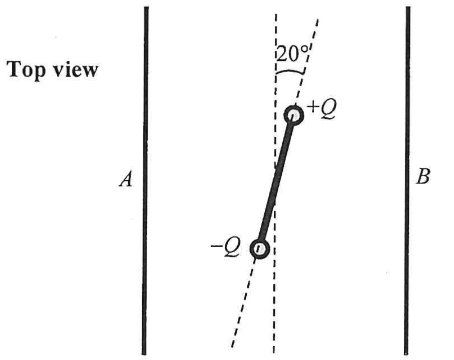
9. Two small metal spheres are attached to the ends of an insulating rod of length 5.0 cm. They carry charges $+Q$ and $-Q$ respectively of equal magnitude as shown in Figure 9.1. The insulating rod is suspended horizontally between two parallel metal plates, A and B, which are connected to an E.H.T. (extra high tension) supply.

Figure 9.1



The rod is parallel to the metal plates when the E.H.T. is off. After the E.H.T. is switched on, an electric field is set up between the plates and the rod is twisted by an angle of 20° as shown in Figure 9.2.

Figure 9.2



(a) On Figure 9.2, sketch the electric field lines due to the potential difference across the plates. (2 marks)

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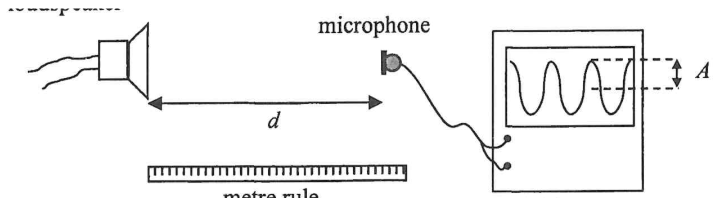

Describe

(I) the procedure of the experiment;

(II) how the data should be analysed using a graphical method to verify whether the hypothesis ($A \propto \frac{1}{d^2}$) is valid.

(5 marks)